# Comparison of Eye-Tracking Representations for Engagement Inference in Office Workers

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## Abstract

## Keywords

eye tracking, engagement inference, representation learning, GazeMAE, MOMENT, subject generalisation

## 1 Introduction

- Engagement inference in office work
- Eye tracking: gaze behaviour, pupil dynamics, behavioural evidence
- Practical representation choice for eye-tracking windows
- Raw temporal signals: information retention, high dimensionality
- Handcrafted features: domain knowledge, interpretability
- Learned embeddings: transfer promise, preprocessing and deployment cost
- Gaze-specific GazeMAE; general time-series MOMENT
- Evidence gap: like-for-like comparison on identical windows and labels
- Unseen-person generalisation; subject-disjoint evaluation
- Research question: representation reliability for self-reported engagement in unseen office workers
- Controlled comparison: raw, handcrafted, GazeMAE, MOMENT
- Frozen pretrained encoders; trained prediction head only
- Contribution: leakage-safe LOSO evaluation
- Contribution: practical representation-selection guidance

## 2 Related Work

- Eye tracking for engagement, cognitive load, affect, and user-state inference
- Gaze and pupil feature engineering
- Gaze representation learning; GazeMAE [1]
- General-purpose time-series foundation models; MOMENT [2]
- Frozen encoder transfer; out-of-the-box downstream utility
- Subject effects and overoptimistic random window splits
- Subject-disjoint evaluation and fold-local preprocessing
- Study position: empirical representation comparison; no new encoder or fine-tuning method

## 3 Methodology

### 3.1 Data and Eye-Tracking Representations

- TrustMe study; office-worker recordings
- Participant count, sessions, recording duration, sampling rate, exclusions
- Tobii channels: left/right pupil size, gaze $x/y$, average distance
- q5 engagement item: “How immersed in your work did you feel just now?”
- q5 response scale and observed raw-label distribution
- Window duration, overlap, temporal ordering
- Common `window_uid` intersection across all four representations
- Labelled windows only; final subject and window counts
- Validity fields; missing-value and quality-control policy
- Display-resolution metadata; per-recording $x/y$ normalization to $[0, 1]$
- Out-of-screen coordinates: retained for modelling policy; excluded from descriptive heatmap
- Representation order: raw, handcrafted features, GazeMAE embeddings, MOMENT embeddings
- Raw: five channels, 120 samples per channel, truncate/pad policy, final dimensionality
- Handcrafted: feature families, metadata removal, missingness filtering, correlation filtering
- GazeMAE: checkpoint, inputs, embedding selection, dimensionality, frozen encoder
- MOMENT: checkpoint, inputs, pooling, dimensionality, frozen encoder
- **Figure 1 — normalized gaze-density map:** screen-coordinate coverage; recording-quality context; non-predictive descriptive evidence

### 3.2 Experimental Design

- Primary task: binary q5 engagement inference
- LOSO target construction: training-subject mean centring; held-out subject global-training-mean centring
- Diagnostic visualization label only: raw q5; binary q5 above global processed-data median; no use in model training or metrics
- Same aligned cohort, target definition, and folds across representations
- Majority-class baseline
- Random Forest: 300 trees, balanced class weights, fixed seed
- MLP: GELU, layer normalization, Adam, early stopping, fixed seed
- Representation-by-model grid; robustness check rather than model-comparison claim
- Fold-local handcrafted-feature selection and imputation
- Fold-local median imputation; train-fit standardization; held-out-subject transform
- No test-subject information in preprocessing parameters
- Accuracy: primary metric


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- Accuracy delta and balanced-accuracy delta: paired majority-baseline comparison within LOSO folds
- Macro-F1 and ROC-AUC: secondary discrimination metrics
- Subject-macro aggregation
- One-class test folds: retained accuracy; undefined balanced accuracy, macro-F1, and AUC; reported valid-fold counts
- Persistence baseline: contextual within-subject temporal check; text only; not comparable with unseen-subject main table

## 4 Results

- **Table 1 — primary LOSO results:** majority baseline; raw, handcrafted, GazeMAE, MOMENT; Random Forest and MLP
- Table columns: representation; model; accuracy (Δ majority); balanced accuracy (Δ majority); macro-F1; ROC-AUC
- Table order: majority baseline; raw; handcrafted; GazeMAE; MOMENT
- Subject-macro estimates; valid-fold counts in caption or table note
- Bold: best value per metric; no overemphasis on small model differences
- Main result: highest accuracy; absolute and majority-baseline improvement
- Representation result: raw and handcrafted performance relative to frozen GazeMAE and MOMENT embeddings
- Model result: consistency of representation ranking across Random Forest and MLP
- One-class held-out-subject treatment; metric availability
- Persistence-check result: contextual temporal predictability; separate protocol; no direct ranking against LOSO models
- **Figure 2 — selected projection:** one representation-method combination; raw q5 continuous colour scale
- Figure-selection criterion: clearest engagement gradient or clearest absence of label separation
- Figure interpretation: exploratory geometry; mixed colours as limited label separability; no causal or performance claim

## 5 Conclusion

- Discussion part:
- Direct answer to the research question
- Frozen learned embeddings: out-of-the-box utility relative to raw and handcrafted alternatives
- Representation effect; secondary role of classifier choice
- Practical recommendation: accuracy, preprocessing burden, storage, computation, interpretability
- Implication of non-improving embeddings: task–pretraining mismatch; need for domain adaptation or fine-tuning
- Projection and heatmap: descriptive support only; no separability proof
- Persistence result: temporal autocorrelation; distinct from cross-person generalisation
- Limitations: single dataset, office-worker sample, self-report noise, binary target construction
- Limitations: frozen checkpoints, fixed windowing, quality-control choices, limited subject count
- Future work: multi-dataset validation, calibrated personalisation, adaptation/fine-tuning, alternative target formulations
- Conclusion part:
- Controlled four-representation comparison for unseen-worker engagement inference
- Final best-performing representation and headline accuracy
- Frozen-encoder conclusion; out-of-the-box deployment recommendation
- Concise practitioner guidance: representation choice under limited labelled data

## Acknowledgements

## References

[1] Louise Gillian C. Bautista and Jr. Naval Prospero C. 2020. GazeMAE: general representations of eye movements using a micro-macro autoencoder. *arXiv preprint arXiv:2009.02437*. https://arxiv.org/abs/2009.02437.

[2] Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. 2024. MOMENT: a family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*. https://arxiv.org/abs/2402.03885.